MIL-E-18239D(AS)
17 December 1982
SUPERSEDING
MIL-E-18239C(WEP)
15 April 1960

MILITARY SPECIFICATION

EARPHONE H-87B/U

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of moving-coil, dynamic type earphone designated as H-87B/U. This earphone transforms electrical energy into acoustical energy and may be acquired normally as component parts of a complete headset assembly.

1.2 Classification. Earphones shall be of the following types as specified (see 6.2).

Type	Figure	Part Number
Type 1	Figure 1	M18239-1
Type 2	Figure 2	M18239-2
Type 4	Figure 3	M18239-4

Type 2 supersedes Type 3, items in stock may be used until exhausted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

GG-C-560 Compass, Magnetic, Surveyor S, Ball Joint and Jacobs Staff, Transit, Pocket

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5965

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SPECIFICATIONS

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-15035	Plastic Sheet, Laminated, Thermosetting, Cotton-fabric-base, Phenolic-resin
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

(Copies of Specifications, Standards, Drawings and Publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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ANSI S1.12-1967 (R1972)	Specifications for Laboratory Standard Microphones
ANSI S3.7-1973	Method for Coupler Calibration of Earphones
USAS S1.10-1966 (R1971)	Method for the Calibration of Microphones

(Copies may be obtained by writing to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Earphones furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for opening of bids (see 4. and 6.).

3.2 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400, Class I for design and construction unless otherwise specified herein.

3.3 Insulating and case materials. Insulating and case materials shall conform to the following requirements, as applicable.

3.3.1 Laminated plastic material. Laminated plastic material shall be in accordance with MIL-P-15035, type FBE.

3.3.2 Molded plastic material. Molded plastic material shall be in accordance with MIL-M-14. Type CFG shall be used where electrical properties are predominant. Type CFI-5 shall be used where mechanical properties are predominant.

3.3.3 Case materials. Case materials shall be of steel, flame retardent modified polyphenylene oxide or other suitable magnetic shielding material. Earphone shall have a jet black finish except the front magnetic shield. The front magnetic shield shall be protected against corrosion by ample plating in accordance with applicable specifications, if applicable.

3.3.4 Insulating and impregnating compounds. All such compounds including varnishes, waxes, etc., shall be suitable in every respect for each particular application. Under all service conditions and test conditions specified in section 4 of this specification, the compound shall adequately preserve the dielectric strength of the insulation to which it is applied. A compound shall not, either in the state of

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its original application or as a result of aging under severe service conditions, show any injurious effect upon the insulation it is designed to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.3.5 Magnetized materials. The materials used in the magnetic circuit shall be of such character and quality, and shall be so processed and assembled, that the earphone will not suffer seriously objectionable degradation in performance, because of loss of magnetization, over long periods of service use or storage.

3.3.6 Cementing compounds. Any cementing compound used in the construction of the earphone shall be of such character and quality that it will withstand all service and test conditions outlined in section 4 without evidence of loosening or otherwise affecting the performance requirements of the earphone.

3.3.7 Moisture and fungus-resistant treatment. Earphones shall be fungusproofed by selection of parts and materials as nonnutrient for fungus or the parts and materials shall be treated prior to their use in the earphones.

3.3.8 Interchangeability. The earphones shall meet the interchangeability requirements as defined in MIL-E-5400.

3.3.9 Weight. The weight of each earphone shall be not greater than 2.5 ounces.

3.3.10 Size. Dimensions specified in Figures 1, 2 and 3, including overall size contours, holes, hole locations, etc., shall be met.

3.3.11 Surface treatment and finishing.

3.3.11.1 Surface treatment. Surface treatment of metal parts shall be in accordance with MIL-S-5002.

3.3.11.2 Finishing. Finishing shall conform to MIL-F-14072.

3.3.12 Soldering. Only resin, or resin and alcohol, may be used as a flux in the assembly of this equipment or any part thereof, except that an acid flux may be used on all-metal, nonelectrical subassemblies provided that they are thoroughly and completely cleaned of all traces of the acid immediately after the soldering operation has been completed. Soldering shall be in accordance with MIL-S-6872, when applicable.

3.3.13 Pressure. The earphones shall be designed such that suitable equalization of the pressure on both sides of the diaphragm shall take place when the earphone is subjected to changes in ambient pressure down to 3.4 inches of mercury.

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3.3.14 Explosive decompression. The earphone unit shall be designed so that the unit will show no structural failure or defect and only minor changes in frequency response characteristics as a result of the explosive decompression test required by 4.7.16.

3.3.15 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4 Performance.

3.4.1 1,000 Hertz sound pressure output. The normal sea level sound pressure vs frequency characteristics of the earphone shall be 103 ± 2 db above a reference level of 0.0002 dyne per square centimeter with 1 milliwatt rms power at 1,000 Hz applied to the earphone terminals in accordance with 4.7.1.

3.4.1.1 Sound pressure output vs frequency. The normal sea level sound pressure vs frequency characteristics of each earphone shall not deviate from the 1,000 Hz response by more than the values shown in Table I when tested in accordance with 4.7.1.

TABLE I. Deviations of sound pressure vs frequency.

Frequency in Hz	Deviations (in db)	
	Minimum	Maximum
100 to 2,500	-3	+3
2,500 to 3,000	-3	+5
3,000 to 7,000	-7	+6

3.4.2 Temperature cycling. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 3 db from normal sea level performance at any frequency between 100 and 7,000 Hz after being subjected to 5 temperature cycles ranging between -55°C and +85°C as specified in 4.7.2.

3.4.2.1 Temperature extremes. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 3 db from normal sea level performance at any frequency between 100 and 7,000 Hz after exposure to temperature of -55°C and +55°C, as required in 4.7.3.

3.4.3 Ambient pressure cycling. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 1 db from normal sea level performance at any frequency between 100 and 7,000 Hz after being subjected to 5 cycles of ambient pressure ranging between 3.4 and 30 inches of mercury as required in 4.7.4.

3.4.3.1 Response at specific ambient pressures. The sound pressures vs frequency characteristics of the earphone unit shall not vary more than 4 db from normal sea level performance between 100 and 1,000 Hz when subjected to an ambient pressure of 13.75 inches of mercury, nor more than 12 db at an ambient pressure of 3.4 inches of mercury when tested as required by 4.7.5.

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3.4.4 Impedance. The impedance of the earphone at 1,000 Hz shall be 15 ± 2 ohms when tested as required by 4.7.6. At any frequency between 100 and 7,000 Hz the impedance shall not vary from the 1,000 Hz value by more than 20 percent when tested as required by 4.7.6.

3.4.5 Insulation resistance. The insulation between the windings and the core, and between the case and terminal shall withstand 500 volts rms 60-cycle ac without breakdown when tested as required by 4.7.7. The insulation resistance shall exceed 1 megohm between the indicated points following the application of this voltage.

3.4.6 Distortion. The rms harmonic distortion in the acoustic output of the earphone shall not exceed 2.5 percent at any fundamental frequency between 200 and 7,000 Hz, when tested as required by 4.7.8.

3.4.7 Overload. The design of the earphone unit shall be such that the unit will show no structural failure or defect and only minor changes in frequency response characteristics as a result of the overload test required by 4.7.9.

3.5 Environmental.

3.5.1 Humidity. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 6 db from normal sea level performance at any frequency between 100 and 7,000 Hz when tested in accordance with the humidity test of 4.7.10.

3.5.2 Moisture. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 3 db from normal sea level performance at any frequency between 100 and 7,000 Hz following exposure to the moisture resistance test as specified in 4.7.11.

3.5.3 Salt spray. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 3 db from normal sea level performance at any frequency between 100 and 7,000 Hz after drying following exposure to the salt spray test as specified by 4.7.12.

3.5.4 Impact. The sound pressure vs frequency characteristics of the earphone unit shall not vary more than 3 db from normal sea level performance at any frequency between 100 and 7,000 Hz and no structural failure shall be evidenced after 12 drops at random as specified by 4.7.13.

3.5.5 Electromagnetic interference. The electromagnetic interference of the earphone shall cause no more than a 5° deflection of a magnetic compass when tested as specified by 4.7.14.

3.5.6 Aging. The earphone unit shall be such that the unit will show no structural failure or defect and only minor changes (less than 1 db) in frequency response characteristics as a result of the aging test specified in 4.7.15.

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3.6 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.7 Workmanship. The earphone shall be manufactured and finished in a thoroughly workmanlike manner. Workmanship, fabrication, and manufacturing processes shall conform to the requirements of MIL-E-5400. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, plating, staking, riveting, machine screw assemblage, etc. All dimensions and tolerances not specified on the applicable drawings shall be held as close as is consistent with the best shop practice. Where dimensions and tolerances affect the interchangeability, operation, or performance of the earphone, they shall be held or limited accordingly. All units shall be inspected for loose, spattered, or excess solder. Metal chips and other foreign material, if found to be present, shall be removed prior to final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of earphones shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph(s) of this specification.

4.3.1 Inspection system requirements. The inspection system requirements shall be in accordance with MIL-I-45208.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. The qualification inspection sample shall consist of ten items of each manufacturer's part number for which qualification is desired. The qualification inspection shall consist of all the tests of this specification described in 4.7. Contractors not having laboratory

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testing facilities satisfactory to the Government for sampling tests shall engage the services of a commercial testing laboratory acceptable to the Naval Avionics Center (NAC), Indianapolis, Indiana. Items for qualification must be separately packaged and forwarded to the activity responsible for qualification as designated in the letter of authorization. The sample shall in all respects be representative of the manufacturer's production item. Each of the items shall be subjected to the applicable inspections specified in this specification. Samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample for qualification test
Specification
Nomenclature
Military part number
Manufacturer's part number
Name of manufacturer
Submitted (date) under authorization of
(reference letter authorizing the test)

Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance of the finished product shall be upon a lot basis after satisfactory completion of all required tests. Acceptance tests shall consist of Group II and Group III tests in Table I of the detail specification for the specified cable assembly.

4.4.4 Retention of qualification. Contractors wishing to remain on the Qualified Products List shall submit to the preparing activity at intervals no greater than two years, DD Form 1718, Certification of Qualified Products.

4.5 Quality conformance inspections. Quality conformance inspections shall be as specified in Table II.

4.5.1 Inspection lot. An inspection lot shall consist of all earphones produced under essentially the same conditions and presented for inspection and test at the same time. The unit of inspection shall be one earphone.

4.5.2 Group A tests. A random sample shall be chosen from each inspection lot in accordance with Table III-A, Level I of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.0 percent defective. The sample earphones shall be used for all group A tests.

4.5.3 Group B tests. A random sample shall be selected from each inspection lot in accordance with MIL-STD-105, Inspection Level L-6, AQL of 6.5 percent defective. The same sample earphones shall be used for all group B tests.

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TABLE II. Quality conformance inspections.

Inspection	Req. Para.	Test Para.
Group A		
Insulation resistance	3.4.5	4.7.7
Workmanship	3.7	4.7.17
Group B		
Sound pressure output vs frequency	3.4.1.1	4.7.1
Temperature extremes	3.4.2.1	4.7.3
Ambient pressure cycling	3.4.3	4.7.4
Response at specific ambient pressures	3.4.3.1	4.7.5
Impedance	3.4.4	4.7.6
Distortion	3.4.6	4.7.8
Electromagnetic interference	3.5.5	4.7.14
Aging	3.5.6	4.7.15
Group C		
Temperature cycling	3.4.2	4.7.2
Overload	3.4.7	4.7.9
Humidity	3.5.1	4.7.10
Moisture	3.5.2	4.7.11
Salt spray	3.5.3	4.7.12
Impact	3.5.4	4.7.13
Explosive decompression	3.3.14	4.7.16

4.5.4 Group C tests. Four random samples shall be selected from each inspection lot in conformance with MIL-STD-105, Inspection Level L-2, acceptance number of zero. Separate samples shall be used for the overload, temperature cycling and explosive decompression tests. The same sample shall be used for the salt spray and impact tests. The impact test shall be performed after completion of the salt spray test.

4.5.5 Resubmitted inspection lots. The paragraph titled "Resubmitted Lots" of MIL-STD-105 shall apply except that a resubmitted inspection lot shall be inspected by the contractor under supervision of the Government inspector using tightened inspection. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be chosen. Before resubmitting, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the inspection lot shall be furnished by the contractor to the cognizant contracting office.

4.6 Test conditions. All tests specified herein shall be made under the following ambient conditions except as otherwise required in specific test requirements.

a. Temperature - 25°C + 15°C or -5°C
b. Pressure - 28 to 30 inches of mercury
c. Humidity - 15 to 90 percent relative

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4.6.1 Tolerance. Where specific tolerances are not given, the maximum allowable tolerances on test condition measurements shall be as follows:

a. Temperature - ±2.0°C
b. Altitude - ±5 percent in feet
c. Humidity - ±5 percent relative
d. Vibration amplitude - ±5 percent. This tolerance is applicable only to the amplitude measuring instruments.
e. Vibration frequency - ±2 percent. This tolerance is applicable only to the frequency measuring instruments.

4.6.2 Test equipment. All instruments used in performing the measurements required by this specification shall be subject to the approval of the acquiring activity.

4.6.2.1 Special requirement for sound pressure output test equipment. All tests for sound pressure output vs frequency characteristics shall be made using test equipment in accordance with ANSI S1.12-1967 (R1972), ANSI 53.7-1973 and USAS S1.10-1966 (R1971) or as directed by the acquiring activity.

4.6.2.2 Microphone. The standard microphone used for making sound pressure output vs frequency tests shall be in accordance with ANSI S1.12-1967 (R1972). The standard microphone shall be calibrated in accordance with USAS S1.10-1966 (R1971) within 3 months prior to the start of each contract covered by this specification and at least every 6 months during each contract.

4.6.2.3 Audio oscillator. The audio oscillator shall have a frequency range of at least 100 to 10,000 Hz, and shall have a high degree of stability in both output voltage and frequency and shall have a waveform distortion of less than 0.5%.

4.6.2.4 Voltmeter. The voltmeters used must have flat frequency response (±0.2 db) from at least 100 to 10,000 Hz, and must be capable of measuring voltages from 0.001 volt rms to 10 volts or more.

4.6.2.5 Microphone preamplifier. The microphone preamplifier shall have a flat response (±1 db) over a frequency range of at least 100 to 10,000 Hz, shall have a high degree of stability and shall have distortion of less than 1%.

4.6.2.6 Wave analyzer. The wave analyzer shall have a frequency range of at least 100 to 10,000 Hz, shall have high selectivity and a voltage range from 0.001 to 10 volts.

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4.7 Tests.

4.7.1 Sound pressure output vs frequency. Measurements shall be made as follows using the test apparatus specified and calibrated in accordance with 4.6.2:

a. Connect the earphone under test to the test circuit and mount on the 6 cc (cubic centimeter) coupler as shown in Figure 4. Figure 5 gives information on the 6 cc coupler.

b. Adjust the output from the oscillator to 0.245 volts at 1,000 Hz as measured by the voltmeter, see Figure 2.

c. Measure the output from the calibrated microphone and preamplifier unit as indicated by the reading of the voltmeter (see Figure 2), and convert this reading to the equivalent db value above 0.0002 dyne per square centimeter using the most recent available calibration curve for the test microphone in use.

d. Repeat the operations set forth in 4.7.1b and 4.7.1c for each of the following frequencies:

(1) From 100 Hz to 1,000 Hz inclusive, measure at each increment of 100 Hz.

(2) From 1,250 Hz, to 7,000 Hz inclusive, measure at each increment of 250 Hz.

e. The results obtained in 4.7.1c and 4.7.1d shall be plotted on semilog graph paper before forwarding to the acquiring activity. Use an ordinate scale reading from 80 to 140 db above 0.0002 dyne per square centimeter and an abscissa scale reading from 100 to 10,000 Hz, inclusive.

4.7.2 Temperature cycling. The ambient temperature cycling shall be in accordance with MIL-STD-202, method 107, using test condition A of Table 107-2.

4.7.3 Temperature extremes. The sound pressure output vs frequency characteristics of the earphone under test shall be measured using the procedures set forth in 4.7.1 with the earphone subjected to the ambient temperature specified in 3.4.2.1 for at least 1 hour. At the contractor's option, the tests specified may be made at room temperature following exposure of at least 1 hour to the specified temperature extremes provided that all tests are completed within 10 minutes after removal from the temperature chamber.

4.7.4 Ambient pressure cycling. Ambient pressure cycling shall consist of varying the ambient pressure surrounding the earphone under test as follows:

a. Reduce the ambient pressure from normal room pressure to 3.4 inches of mercury at the greatest practical rate of change.

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b. Retain the ambient pressure at between 3.0 and 3.4 inches of mercury for 5 minutes.

c. Increase the ambient pressure to normal room pressure at a rate of from 0.45 to 0.55 inch of mercury per second.

d. Retain the ambient pressure at normal room pressure for 5 minutes.

e. Repeat the procedure set forth in 4.7.4a to 4.7.4d inclusive, until the earphone has been subjected to 5 complete cycles.

f. Measure the sound pressure output vs frequency characteristics of the earphone as described in 4.7.1.

4.7.5 Response at specific ambient pressures. Measure the sound pressure output vs frequency characteristics of the earphone under test using the procedures set forth in 4.7.1 with the earphone subjected to the ambient pressures specified in 3.4.3.

4.7.6 Impedance. Impedance measurements shall be made at 1,000 Hz with the earphones mounted on the 6 cc coupler as shown on Figure 2, using a voltmeter-ammeter circuit satisfactory to the cognizant Government inspector. The 1,000 Hz voltage applied across the terminals of the earphone during impedance measurements shall be 0.245 volts rms.

4.7.7 Insulation resistance. The insulation resistance of each earphone shall be measured by one insulated terminal to the clamping ring or other metallic surface not covered by a nonconducting coating which has good contact to the case of the headphone using any standard ohmmeter circuit which is designed to apply at least 500 volts dc across the test terminals during the measurement. A 500 volt megger resistance test, may be substituted for the 500 volt dc test described above, at the option of the contractor.

4.7.8 Distortion. Measurements shall be made, using the test apparatus specified and calibrated in accordance with 4.6.1, as follows:

a. Connect the earphone under test to the test circuit and mount on the 6 cc coupler as shown in Figure 4.

b. Adjust the output from the oscillator to 2.12 volts rms at 200 Hz as measured by the voltmeter (see Figure 4).

c. With the wave analyzer input connected across the voltmeter (see Figure 2) and operated according to manufacturer's instructions, measure the rms harmonic distortion as computed from the following formula:

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$$\text{Percent harmonic distortion} = \sqrt{\frac{P_2^2 + P_3^2 + \cdots}{P_1^2}} \times 100$$

where:

P_1 is the sound pressure output at the oscillator (fundamental) frequency.

P_2, P_3, etc., are the sound pressure outputs of each respective harmonic frequency having measured ouptut in excess of 10 percent of the output at the fundamental frequency.

d. Repeat the operations set forth in 4.7.8b and 4.7.8c using an oscillator frequency of 400 Hz and such additional frequencies between 400 and 7,000 Hz as are required to determine at what frequency in this range maximum distortion exists.

4.7.9 Overload. Audio power of 500 milliwatts rms at 1,000 Hz shall be applied to each earphone under test for 8 hours. During the last hour of test, the earphone shall be checked for audible rattles or other signs of failure. Upon completion of the overload period, the sound pressure output vs frequency characteristics shall be measured as set forth in 4.7.1.

4.7.10 Humidity. The earphone shall be exposed to 95 percent relative humidity at 50°C (122°F) for 24 hours and shall not vary more than 3 db after drying.

4.7.11 Moisture. Each earphone under this test shall be subjected to moisture resistance test conditions as outlined in MIL-STD-202, method 106.

4.7.12 Salt spray. Each earphone under test shall be subjected to salt spray in accordance with MIL-STD-202, method 101, test condition B. The sound pressure output vs frequency characteristics shall be measured as set forth in 4.7.1.

4.7.13 Impact. Each earphone under test shall be dropped 12 times at random from a height of 3 feet onto a concrete floor or block, then the sound pressure output vs frequency characteristics shall be measured as set forth in 4.7.1.

4.7.14 Electromagnetic interference. Electromagnetic interference measurements shall be made in a shielded room or in an area proved to the satisfaction of the cognizant Government inspector to be free from appreciable magnetic disturbances. The earphone under test shall be rotated on its axis at various points on the surface of an 8 inch radius sphere surrounding a compass which meets the requirements of GG-C-560 type II or equal.

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4.7.15 Resistance to aging. Each earphone under test shall be tested as follows:

a. Place the earphone with its diaphragm in a vertical plane on a standard vibrating machine which has an amplitude of 0.06 inch in such a manner as to provide, individually, motion in three directions (perpendicular, parallel, and 45 degrees to perpendicular) relative to the axis of the earphone which is perpendicular to the diaphragm.

b. Apply audio power of 60 milliwatts rms at 1,000 Hz to the earphone for intervals of 2 minutes on and 2 minutes off.

c. Vary the frequency of the vibrating machine approximately every minute from 10 to 50 to 10 Hz.

d. After a one hour period of the tests described in (b) and (c) above, measure the sound pressure output vs frequency characteristics of the earphone as described in 4.7.1.

4.7.16 Explosive decompression. Each earphone under test shall be subjected to 10 consecutive explosive decompressions from an altitude of 8000 feet to an altitude of 35,000 feet (corresponding to pressures of 22.5 inches of mercury to 7.04 inches of mercury). Each complete decompression or change of pressure shall take place in 0.1 seconds or less. The impedance at 1000 Hz and the sound pressure output vs frequency characteristics shall be measured before and after completion of the test as set forth in 4.7.1.

4.7.17 Workmanship. Each earphone shall be visually inspected to insure that:

a. The earphone sections are tightly clamped or otherwise properly fastened together in accordance with the qualification tested sample.

b. All external finishes, including any plated parts, are free from blemishes which would permit breaking of the protective coating.

c. All markings are complete and in accordance with an approved sample and Figures 1, 2 and 3.

d. The color of the case is in accordance with an approved sample.

e. The weight and all over-all dimensions are correct by comparison with an approved sample or Figures 1, 2 and 3 insertion in an approved fixture.

f. Both cord connector set screws are in place and turn freely in the terminal lug.

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g. The holes in the face of the unit are cut clean, properly centered and are free from foreign materials of any kind.

5. PACKAGING

5.1 Packaging requirements. All units shall be packaged, packed, and marked for shipment in accordance with MIL-P-17555. (The level of preservation, packaging, and packing shall be specified by the acquiring activity.)

6. NOTES

6.1 Intended use. These earphones are intended for use with a variety of headsets and helmets required for aircraft communicating facilities in performance of the function intended (see 3.4). In addition, these earphones are mechanically and electrically interchangeable with earphone: P/OH-87A/U but the frequency range has been extended from 6,000 to 7,000 Hz.

6.2 Ordering data. Acquisition should specify the following:

a. Title, number, and date of this specification.

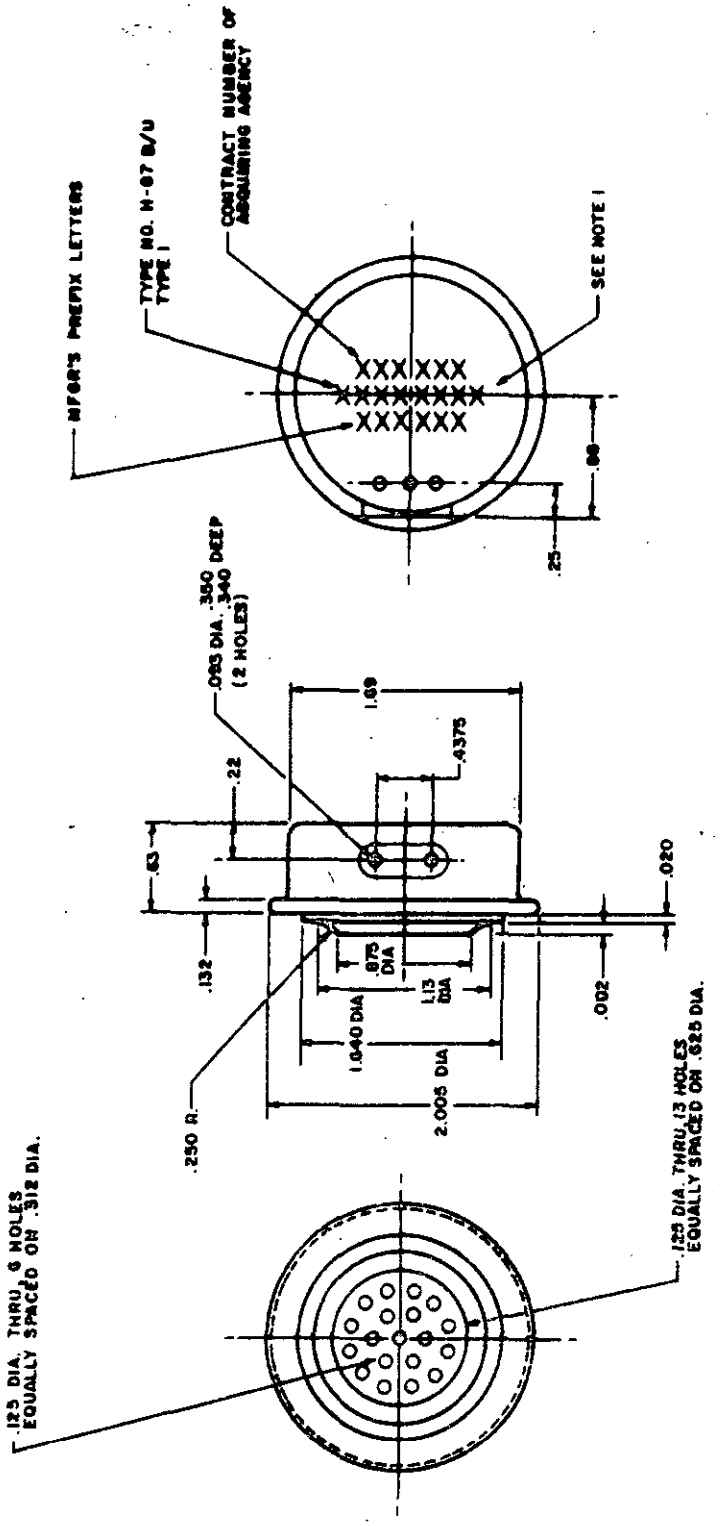
b. Type.

c. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command (Attention: Commanding Officer, Naval Air Engineering Center, (ESSD) Code 93, Lakehurst, NJ 08736); however application for Qualification of Products should be made to Commanding Officer, Naval Avionics Center (Code 432), 6000 East 21st Street, Indianapolis, IN 46218, who has been designated Naval Air Systems Command agent for establishing this Qualified Products List.

Preparing activity:
Navy - AS
(Project No. 5965-N181)

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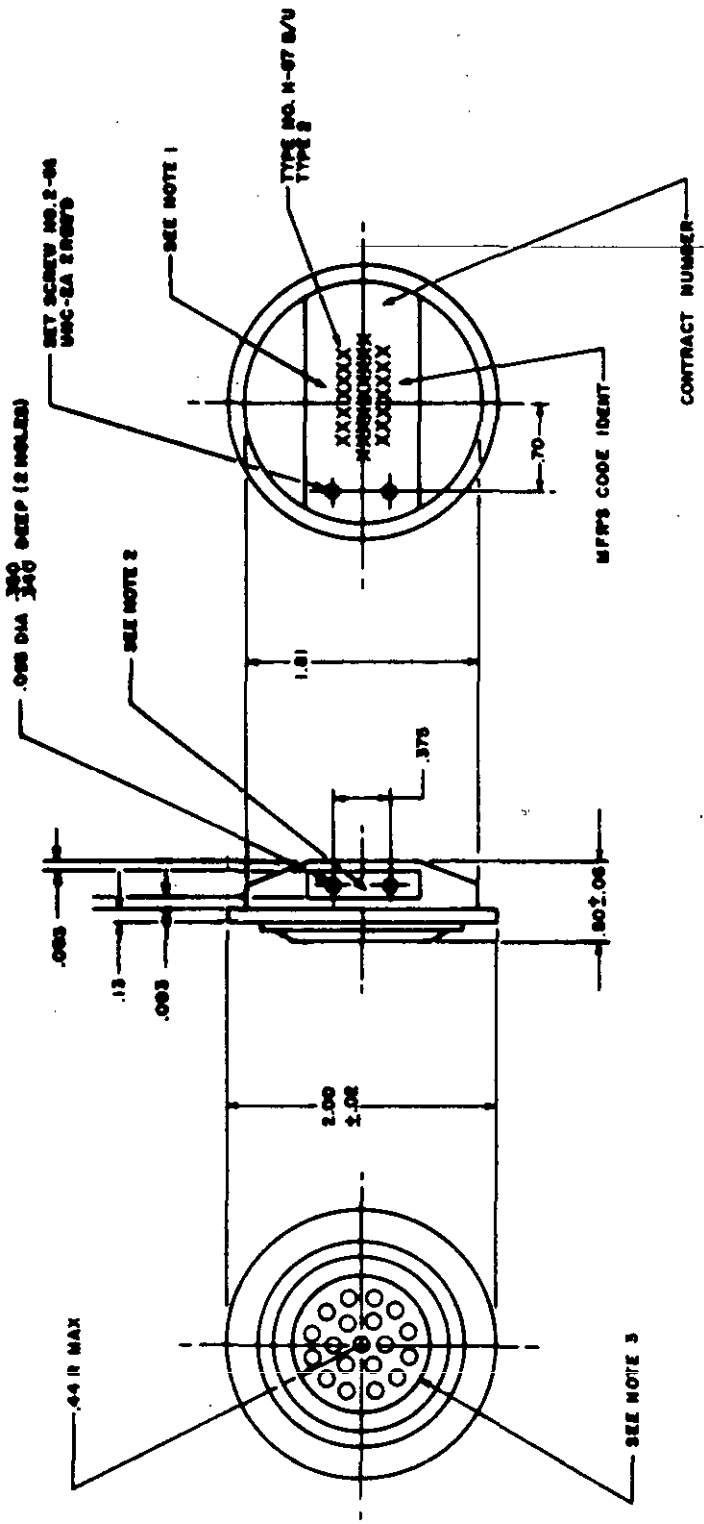


NOTES:

1. CHARACTERS SHALL BE 3/32 INCH HIGH AND SHALL BE IMPRESSED PERMANENTLY.
2. DIMENSIONS ARE IN INCHES. TOLERANCES ARE 2 PLACE DECIMALS ±0.015, 3 PLACE DECIMALS ±0.005 UNLESS OTHERWISE NOTED.

Figure 1. Earphone - Type 1.

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NOTES:

1. CHARACTERS SHALL BE 3/32 INCH HIGH AND IMPRESSED PERMANENTLY INTO THE CASE.
2. CONNECTION HOLES MAY BE LOCATED WITHIN ENVELOPE SHOWN.
3. SIZE, NUMBER AND LOCATION OF HOLES IN CAP TO BE DETERMINED BY MANUFACTURER, SUBJECT TO ACOUSTICAL AND ELECTRICAL REQUIREMENTS AND LIMITING DIMENSIONS.
4. EARPHONE CONFIGURATION MAY VARY FROM THAT SHOWN BUT DIMENSIONS SHALL NOT BE EXCEEDED. MOUNTING DIMENSIONS SHALL BE AS SPECIFIED.
5. DIMENSIONS ARE IN INCHES. TOLERANCES ARE 2 PLACE DECIMALS ±0.010, 3 PLACE DECIMALS ±0.005.

Figure 2. Earphone - Type 2.

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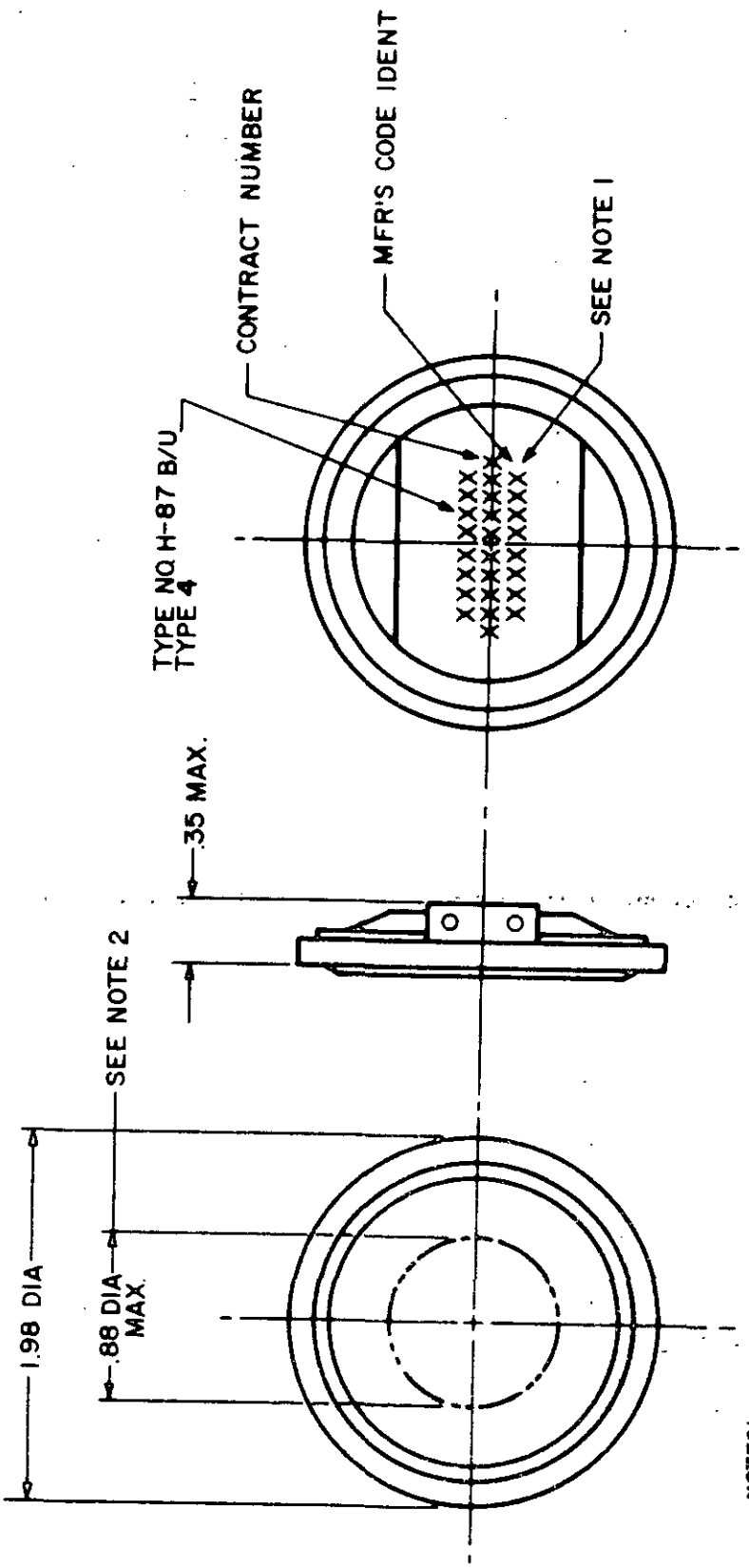
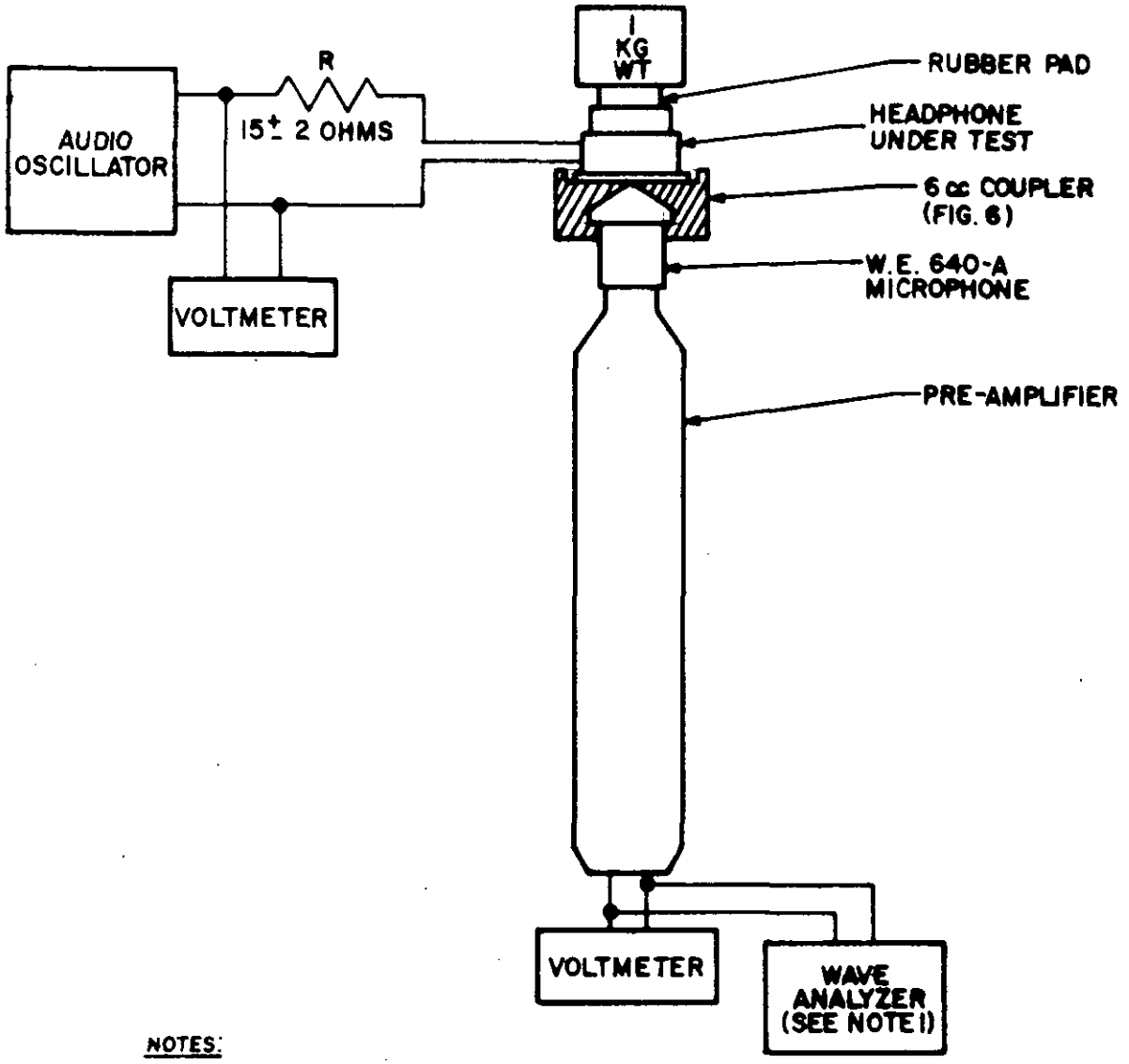


Figure 3. Earphone - Type 4.

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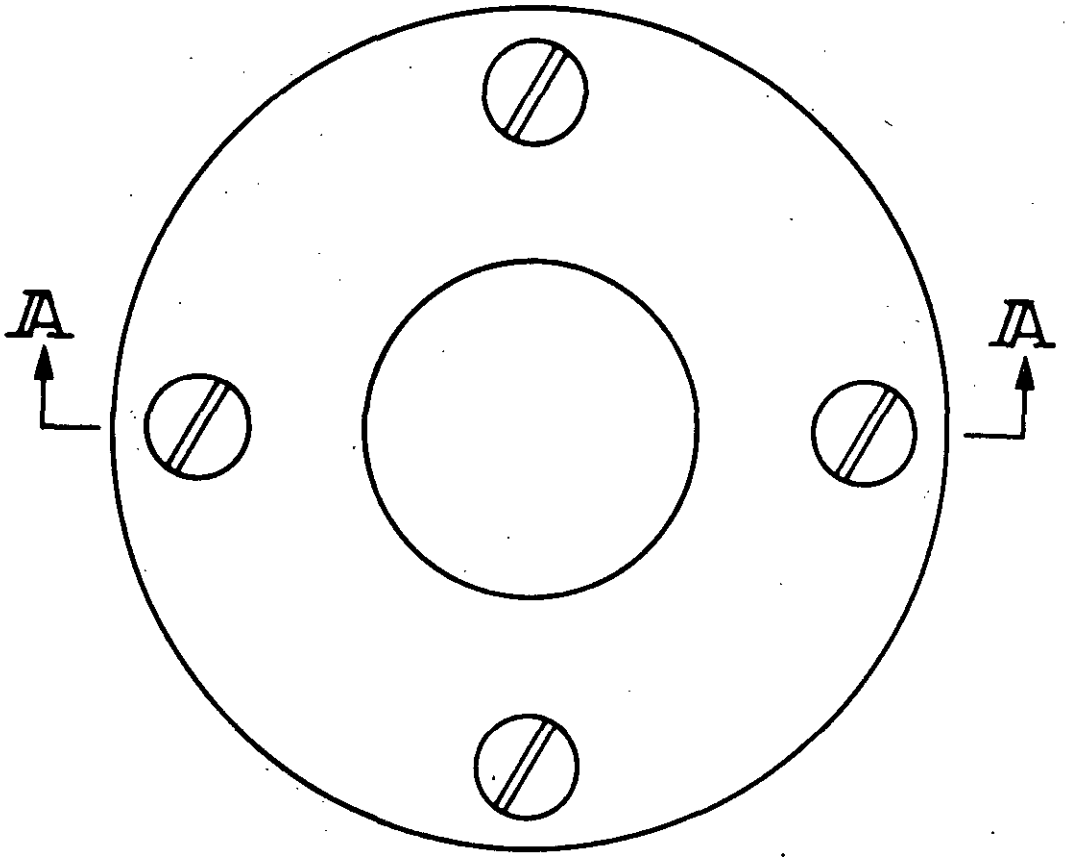


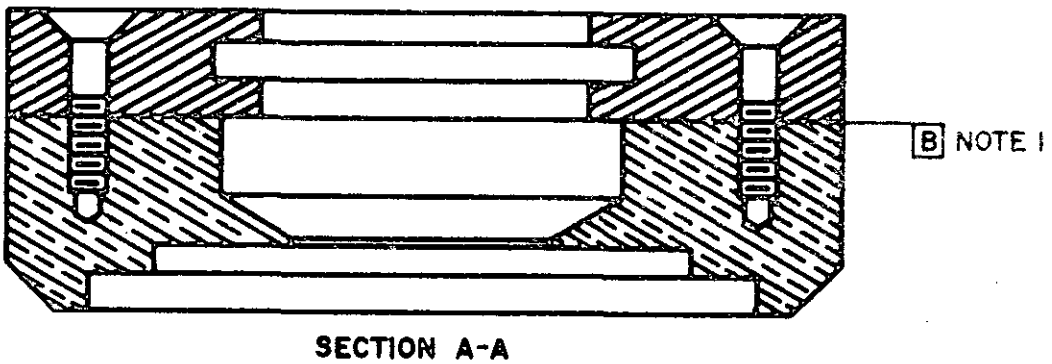
NOTES:

1. WAVE ANALYZER REQUIRED FOR DISTORTION TESTS ONLY.
2. THE ESSENTIAL CHARACTERISTICS FOR THE COMPONENTS SHOWN IN THIS FIGURE ARE SET FORTH UNDER PARAGRAPH 4.5.2 OF THIS SPECIFICATION.

Figure 4. Test circuit.

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SECTION A-A

Figure 5. Coupler, Acoustic 6 cc.

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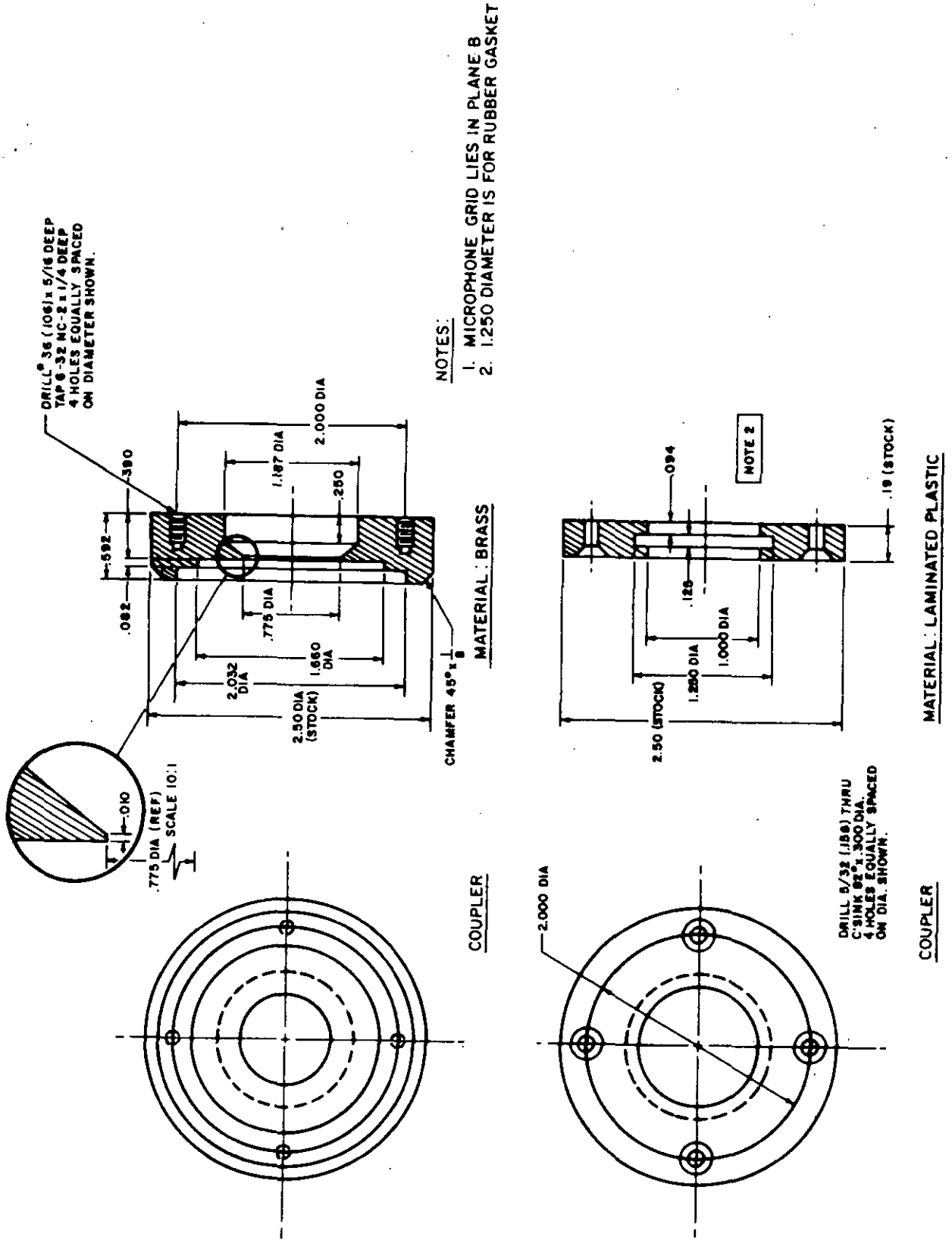


Figure 5. Coupler, acoustic 6 cc.

SHEET 2 OF 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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